

Initial Parameter Configuration.

```
Dropout = 0.1
SEED = 42
layers = 2
optimizer='adam' (default learning rate ~0.001)
loss='mse'
WINDOW=30
Units=32
Epochs=20
Patience=10
batch_size = 32
10 Year Data (5m)
RMSE (Root Mean Squared Error)
MAE (Mean Absolute Error)
```

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 30, 32)	4,352
dropout (Dropout)	(None, 30, 32)	0
lstm_1 (LSTM)	(None, 32)	8,320
dropout_1 (Dropout)	(None, 32)	0
dense (Dense)	(None, 1)	33

Total params: 12,705 (49.63 KB)
Trainable params: 12,705 (49.63 KB)
Non-trainable params: 0 (0.00 B)

SMA 50

```
Epoch 1/20
14029/14029 ----- 207s 15ms/step - loss:
0.0012 - val_loss: 4.6642e-05
Epoch 2/20
14029/14029 ----- 261s 15ms/step - loss:
2.0690e-04 - val_loss: 3.5599e-05
Epoch 3/20
14029/14029 ----- 259s 14ms/step - loss:
1.9790e-04 - val_loss: 1.2728e-04
Epoch 4/20
14029/14029 ----- 204s 15ms/step - loss:
1.9343e-04 - val_loss: 2.8188e-04
Epoch 5/20
14029/14029 ----- 262s 15ms/step - loss:
1.9162e-04 - val_loss: 3.0651e-04
Epoch 6/20
14029/14029 ----- 260s 14ms/step - loss:
1.8656e-04 - val_loss: 9.0392e-04
Epoch 7/20
14029/14029 ----- 262s 14ms/step - loss:
1.8410e-04 - val_loss: 5.9029e-04
Epoch 8/20
14029/14029 ----- 264s 15ms/step - loss:
1.8297e-04 - val_loss: 6.8722e-04
Epoch 9/20
14029/14029 ----- 262s 15ms/step - loss:
1.8216e-04 - val_loss: 4.5816e-04
Epoch 10/20
14029/14029 ----- 263s 15ms/step - loss:
1.7785e-04 - val_loss: 2.4637e-04
Epoch 11/20
14029/14029 ----- 258s 14ms/step - loss:
1.7876e-04 - val_loss: 9.6578e-05
Epoch 12/20
14029/14029 ----- 266s 15ms/step - loss:
1.7807e-04 - val_loss: 4.0013e-04
Epoch 12: early stopping
Restoring model weights from the end of the best epoch: 2.

14029/14029 ----- 57s 4ms/step - loss:
1.4363e-05
MSE Train: 1.598800918145571e-05
4676/4676 ----- 19s 4ms/step - loss:
8.1799e-06
MSE Test: 9.109372513194103e-06
```

